

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 04

Lab Report Name : Connect Two Computers Using Crossover Cable & Share Files

Course Title : Computer Networks Lab

Course Code : ICT-3202

Submitted By	Submitted To
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Date of Performance :

Date of Submission :

Experiment No: 04

Experiment Name: Connect Two Computers Using Crossover Cable & Share Files

Objectives:

1. To understand the functioning of crossover cables and their application in direct computer-to-computer communication
2. To configure IP addresses manually for two computers on the same local network
3. To set up and enable file sharing between two computers connected via crossover cable
4. To implement proper security measures and network profile settings for safe file transfer
5. To demonstrate the practical use of network protocols in local area networking

Necessary Materials and Software:

1. Cat6 Cross (Crossover) Cable
2. Two PCs/Laptops, each with a functional Ethernet (RJ45) network interface card
3. Operating System with networking capability (e.g., Windows 10/11)
4. Command Prompt (ping testing)
5. Control Panel (for IP configuration)
6. File Explorer

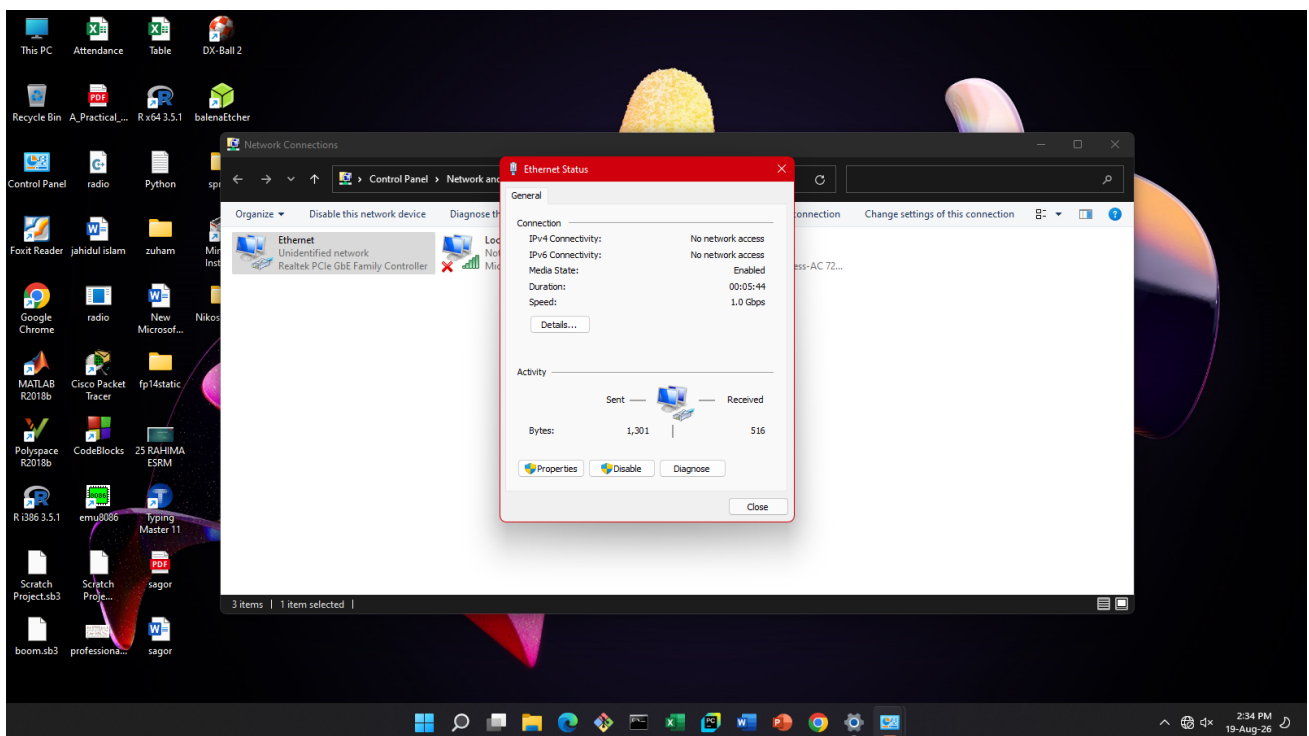
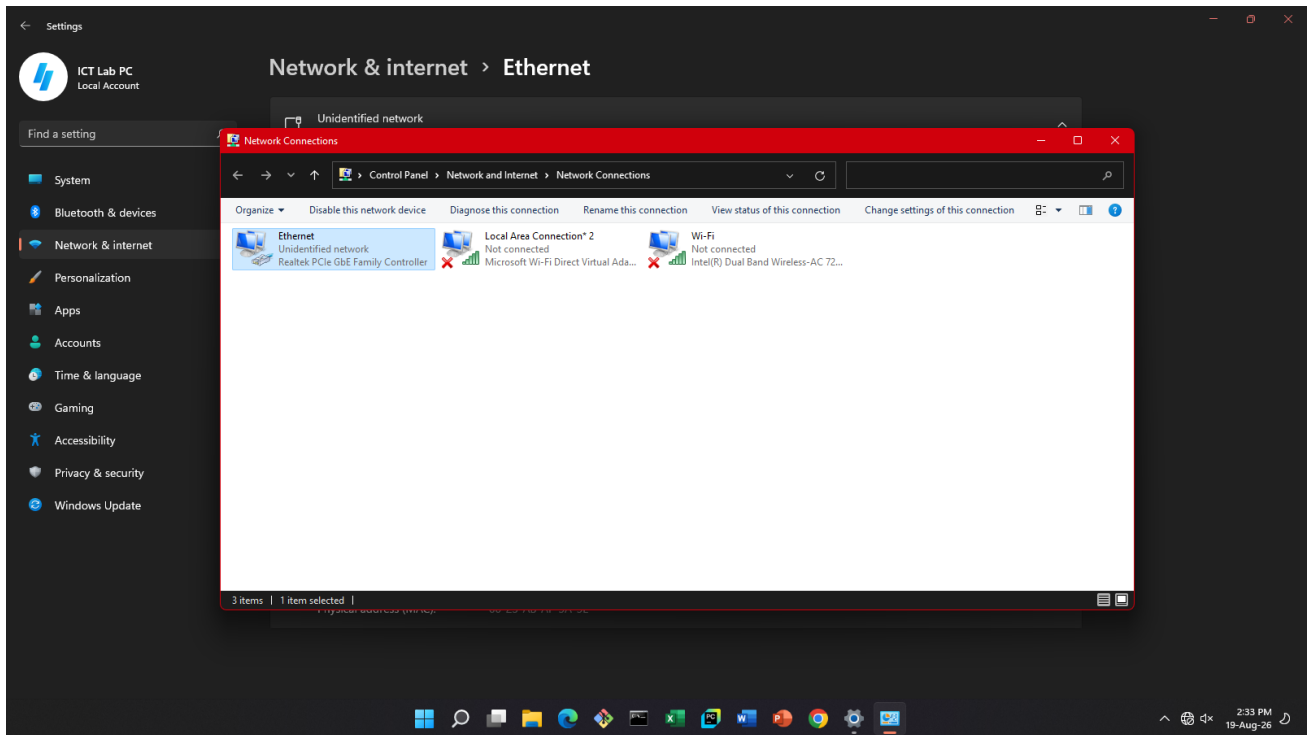
Procedure:

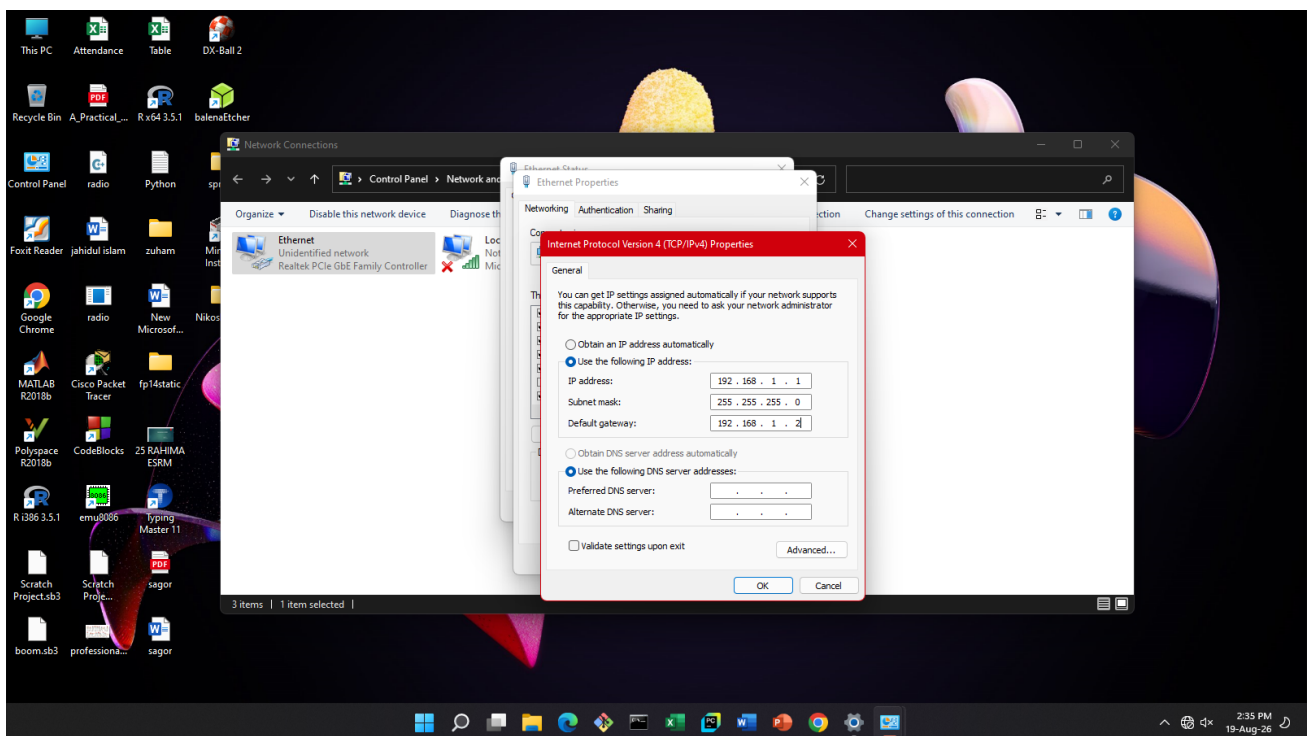
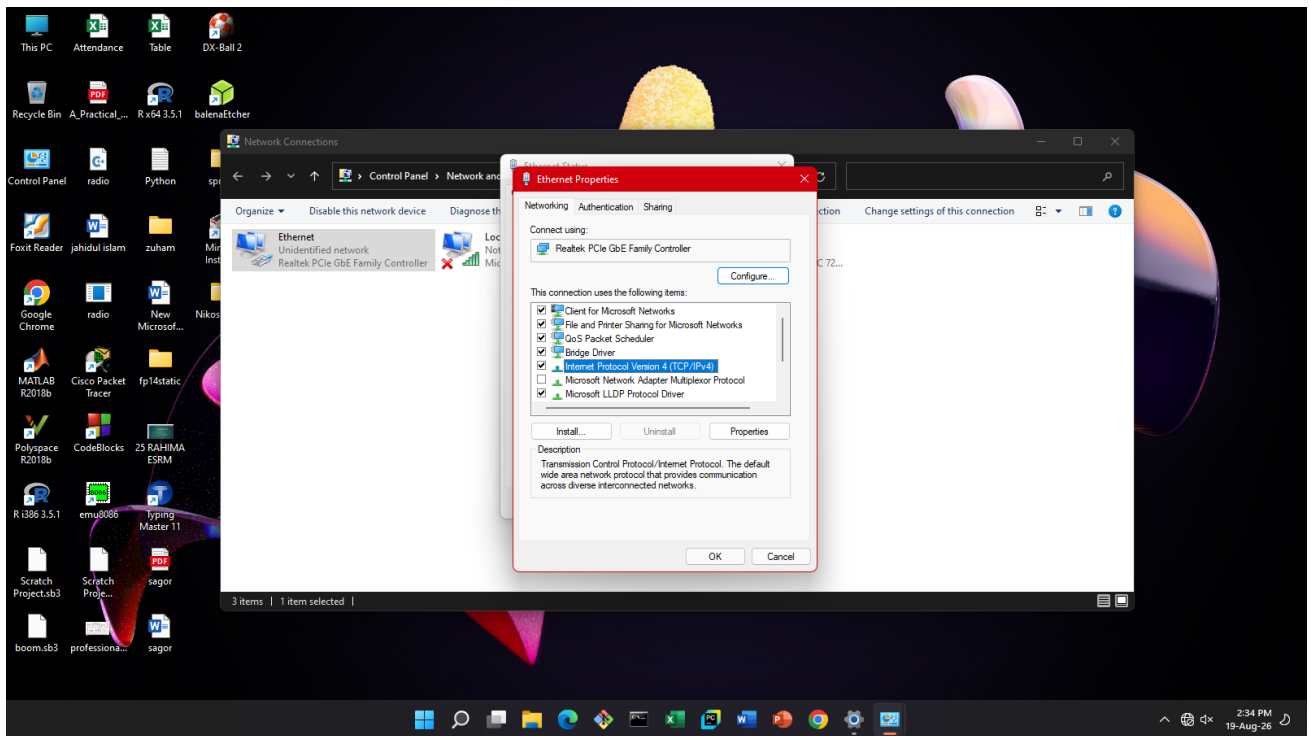
Step 1: Physical Connection

1. Ensure both computers are powered on
2. Plug the crossover cable into the LAN port of Computer 1
3. Plug the other end of the crossover cable into the LAN port of Computer 2
4. Verify that the network LED indicators on both NICs are lit (indicating active connection)

Step 2: IP Configuration for Computer 1

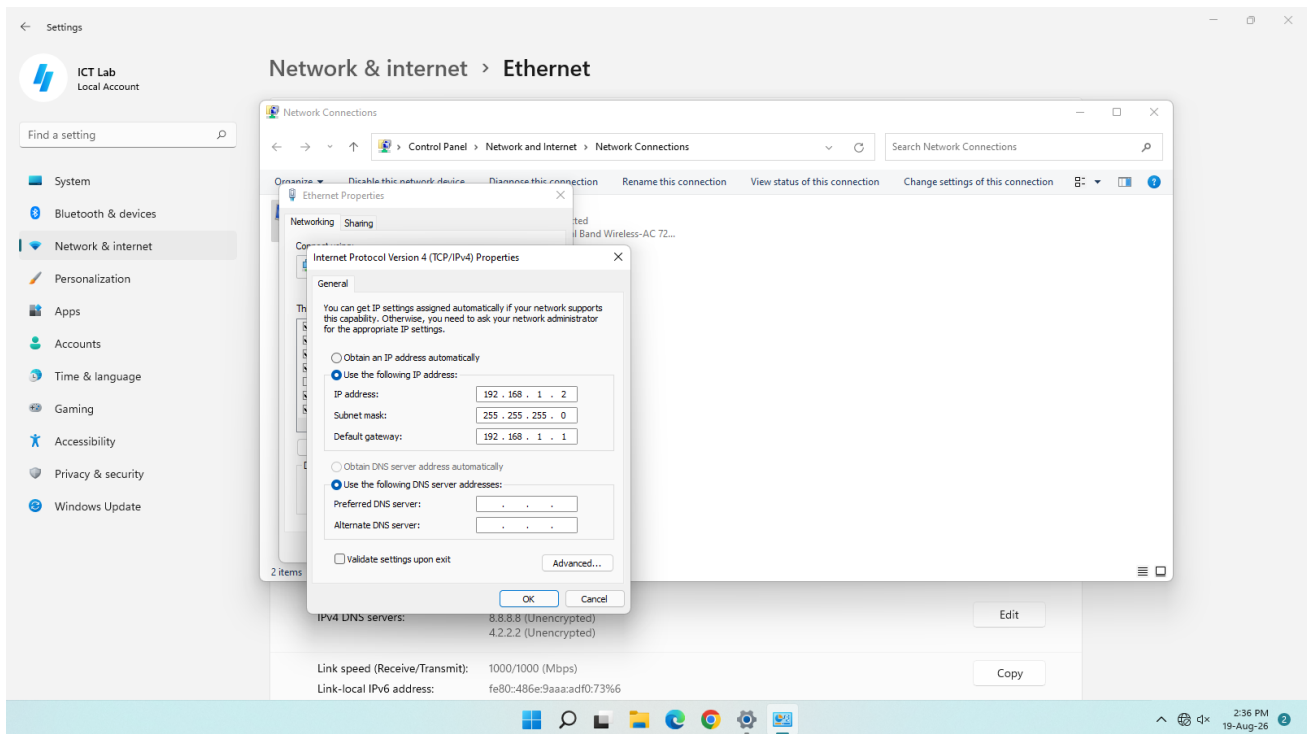
1. Open Settings from the Start menu
2. Navigate to Network & Internet
3. Click on Ethernet
4. Select Properties
5. Click on Internet Protocol Version 4 (TCP/IPv4)
6. Click Properties
7. Configure the following settings:
8. IP Address: 192.168.1.1
9. Subnet Mask: 255.255.255.0
10. Default Gateway: 192.168.1.2
11. Click OK to apply changes





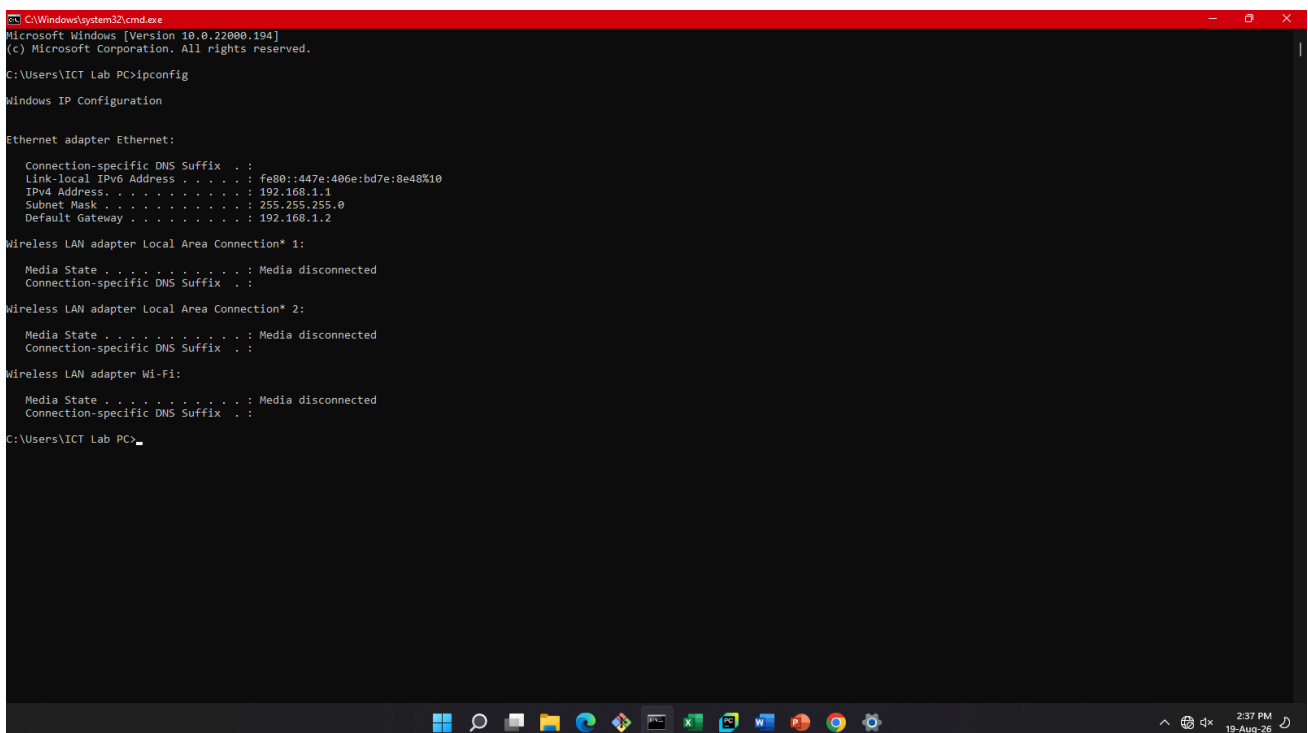
Step 3: IP Configuration for Computer 2

1. Repeat the same process as Step 2 on Computer 2
2. Configure the following settings:
3. IP Address: 192.168.1.2
4. Subnet Mask: 255.255.255.0
5. Default Gateway: 192.168.1.1
6. Click OK to apply changes



Step 4: Verify Network Connectivity

1. Open Command Prompt on Computer 1
2. Use ping command: ping 192.168.1.2
3. If successful, packets should be received from Computer 2
4. Now try on Computer 2 to ping 192.168.1.1 to verify bidirectional communication

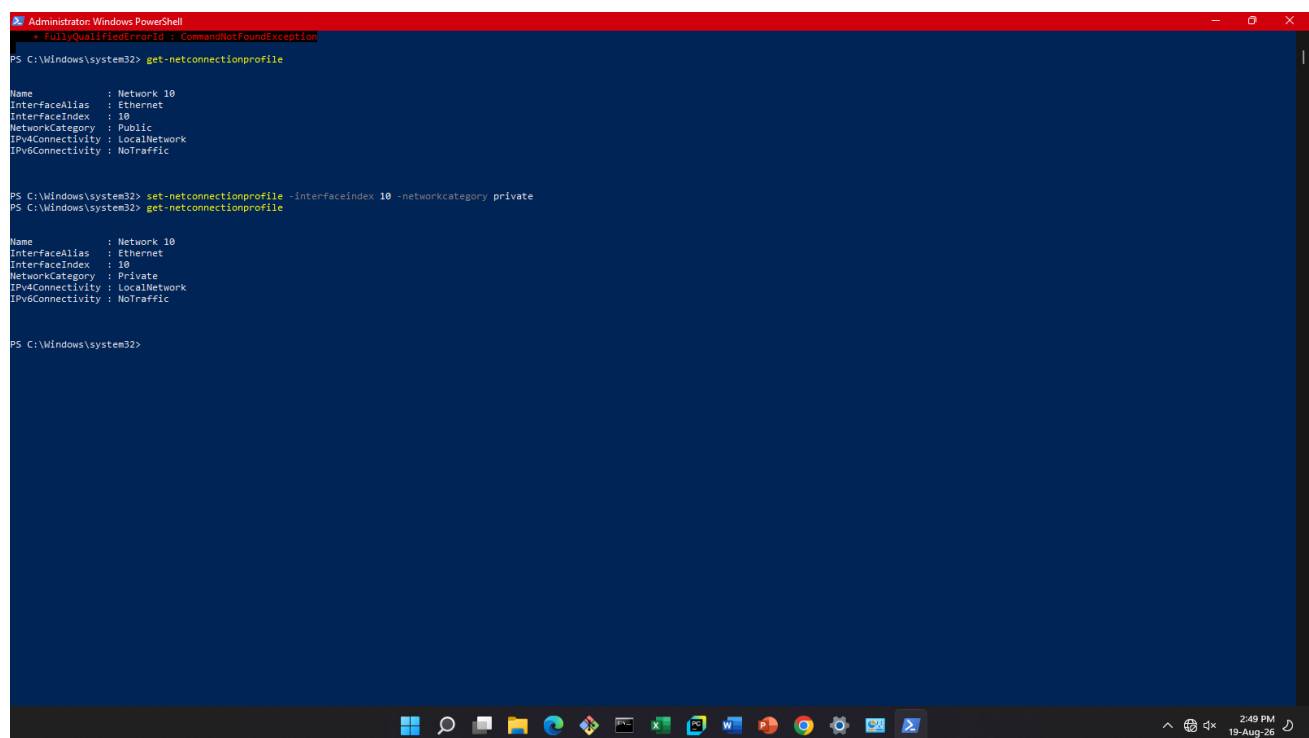


Step 5: Set Network Profile to Private

On Computer 1:

1. Open Settings (Win + I)
2. Click Network & Internet
3. Click on Ethernet (your active connection)
4. Under Network profile type, select Private
5. If using PowerShell: Right-click Start and select Terminal/PowerShell as Administrator
6. Run: `Get-NetConnectionProfile`
7. Run: `Set-NetConnectionProfile -Name "Ethernet" -NetworkCategory Private`

Repeat Step 5 on Computer 2 with the same settings.



```
Administrator Windows PowerShell
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32> get-netconnectionprofile

Name                : Network 10
InterfaceAlias       : Ethernet
InterfaceIndex       : 10
NetworkCategory      : Public
IPv4Connectivity     : LocalNetwork
IPv6Connectivity     : NoTraffic

PS C:\Windows\system32> set-netconnectionprofile -interfaceindex 10 -networkcategory private
PS C:\Windows\system32> get-netconnectionprofile

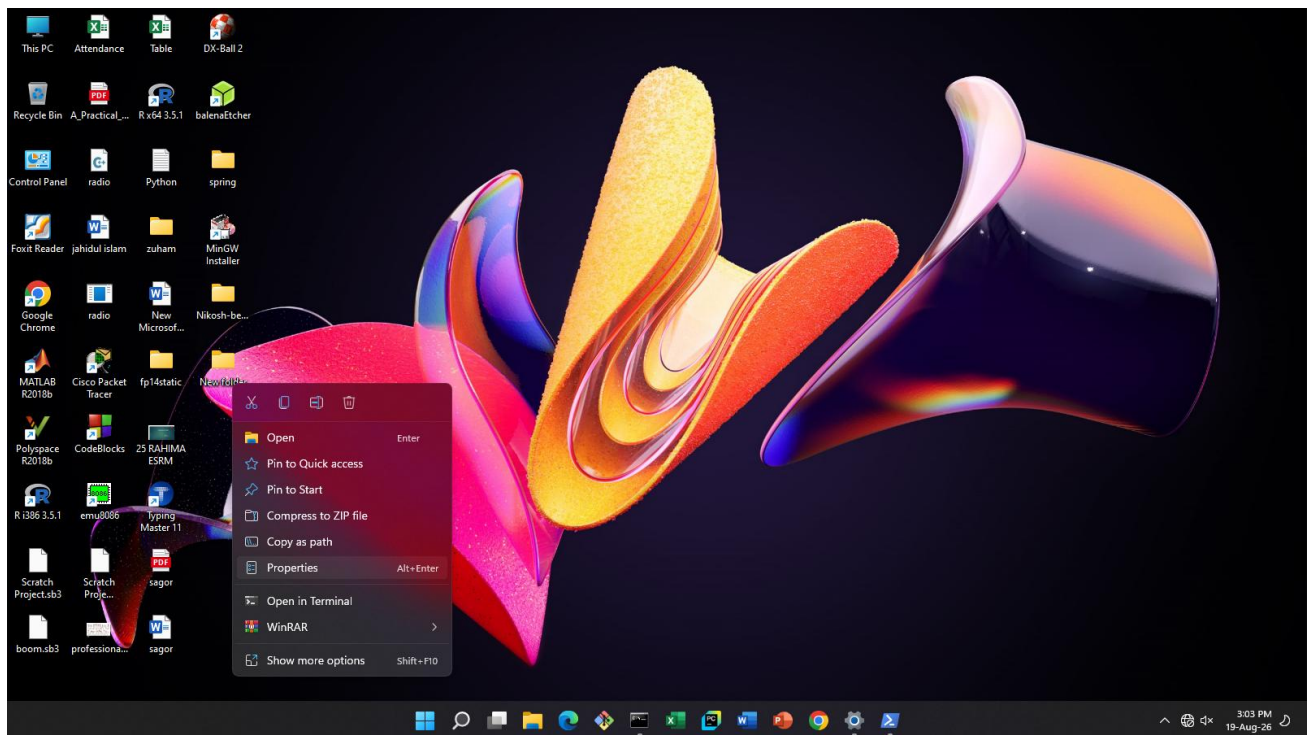
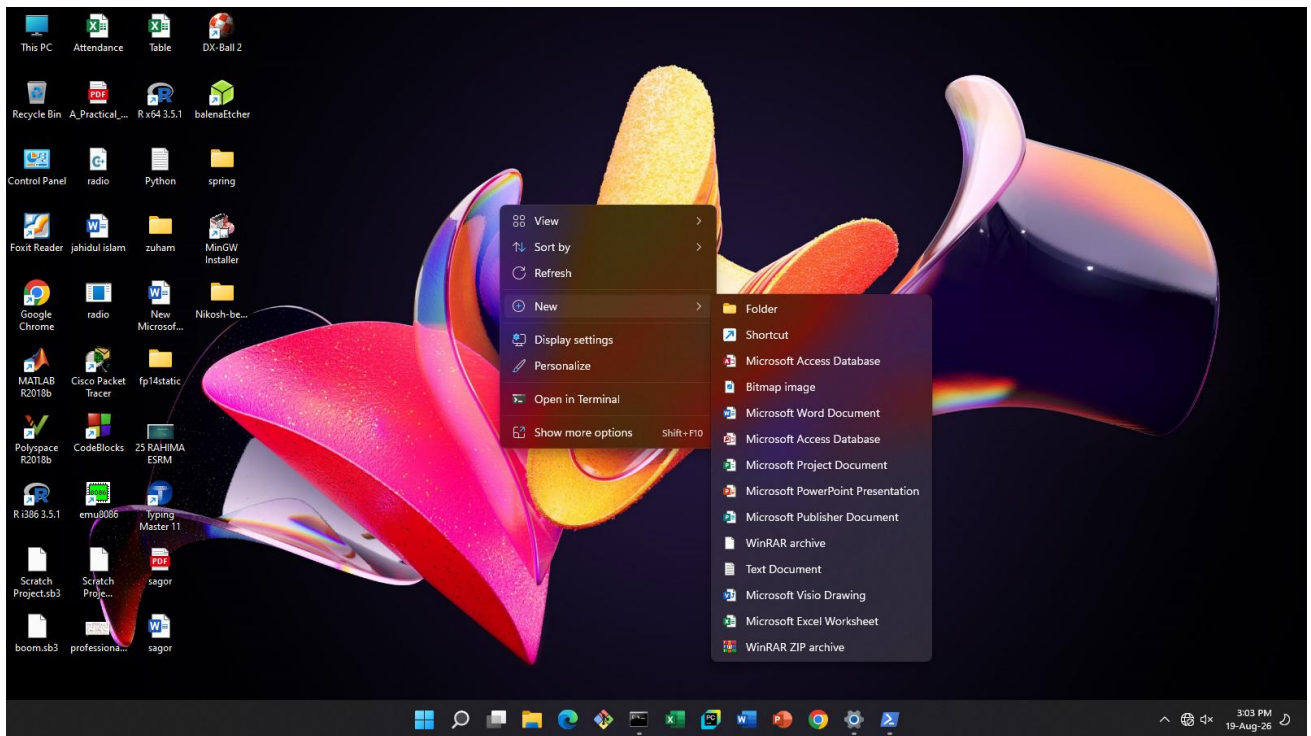
Name                : Network 10
InterfaceAlias       : Ethernet
InterfaceIndex       : 10
NetworkCategory      : Private
IPv4Connectivity     : LocalNetwork
IPv6Connectivity     : NoTraffic

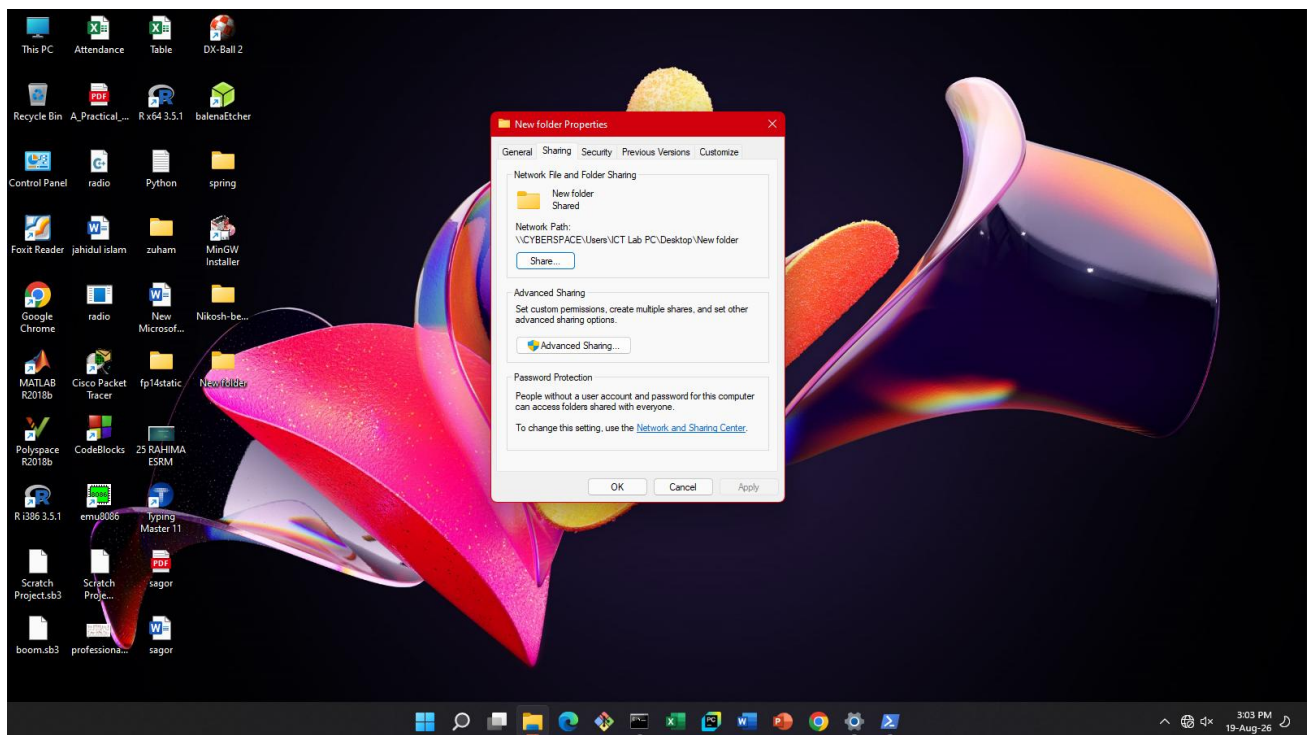
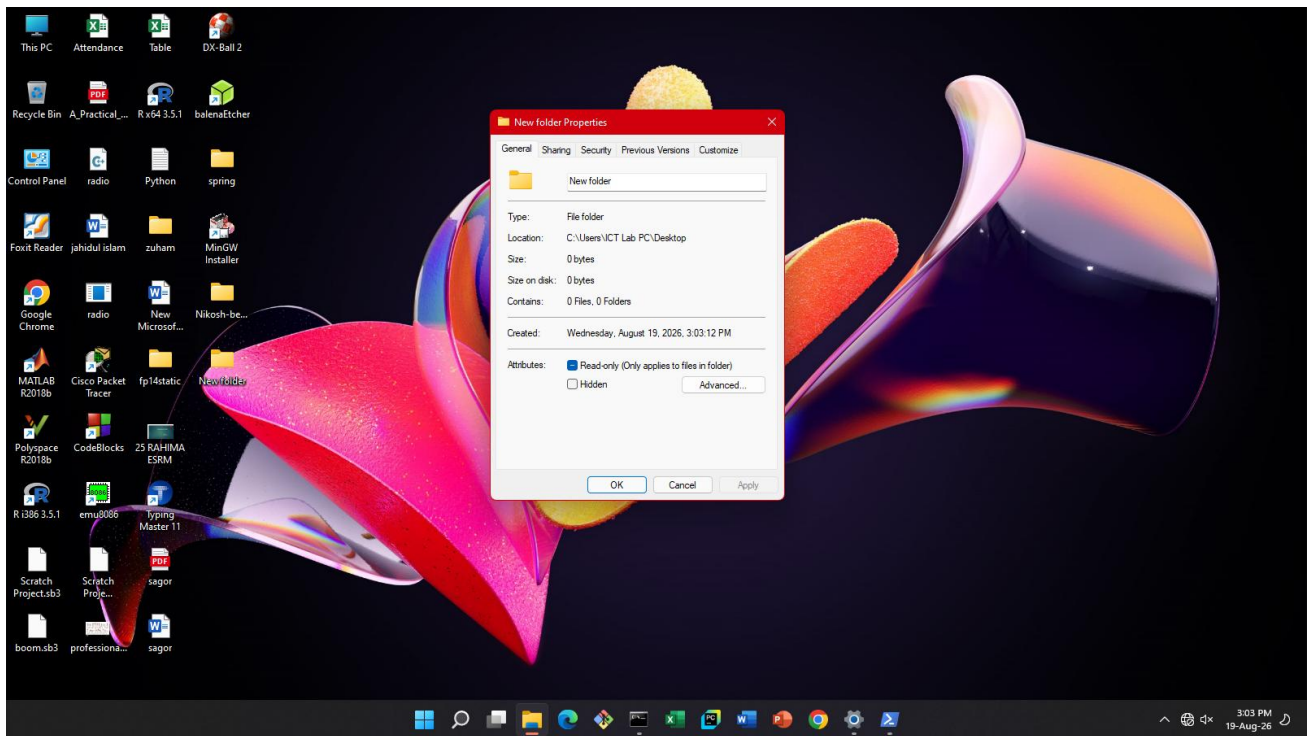
PS C:\Windows\system32>
```

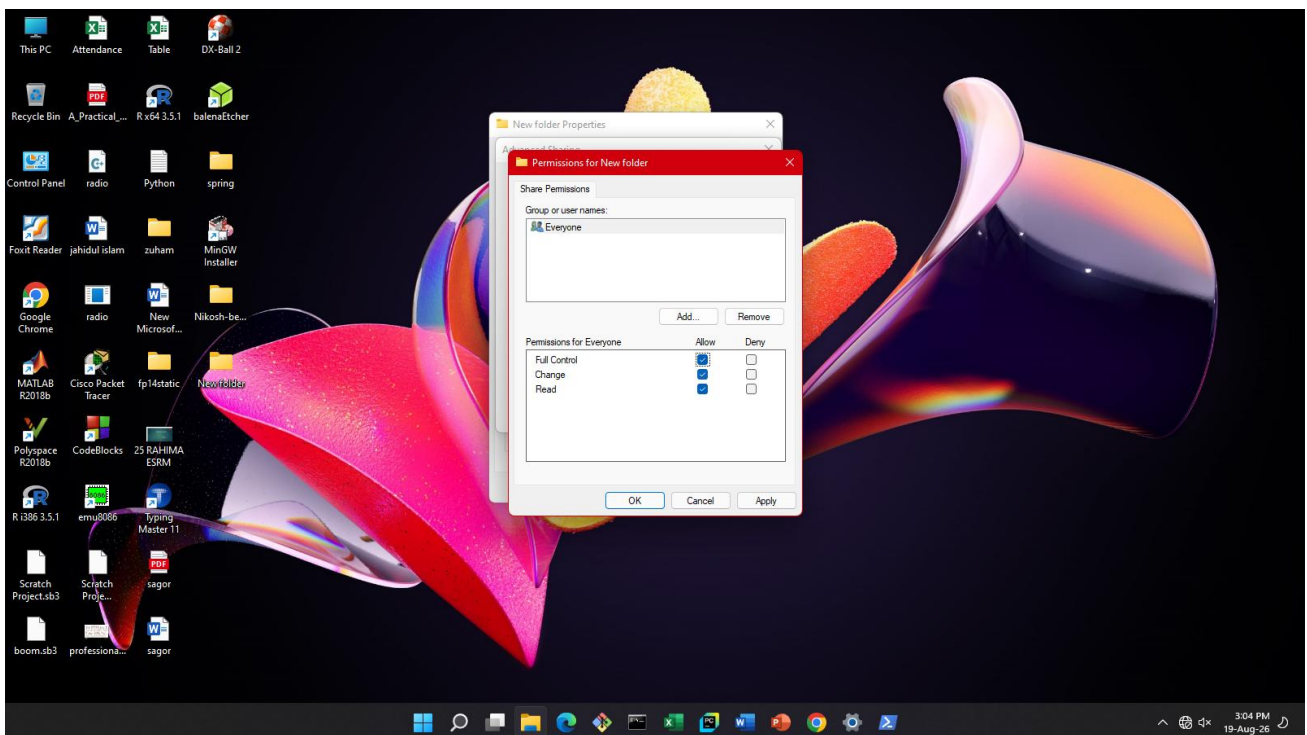
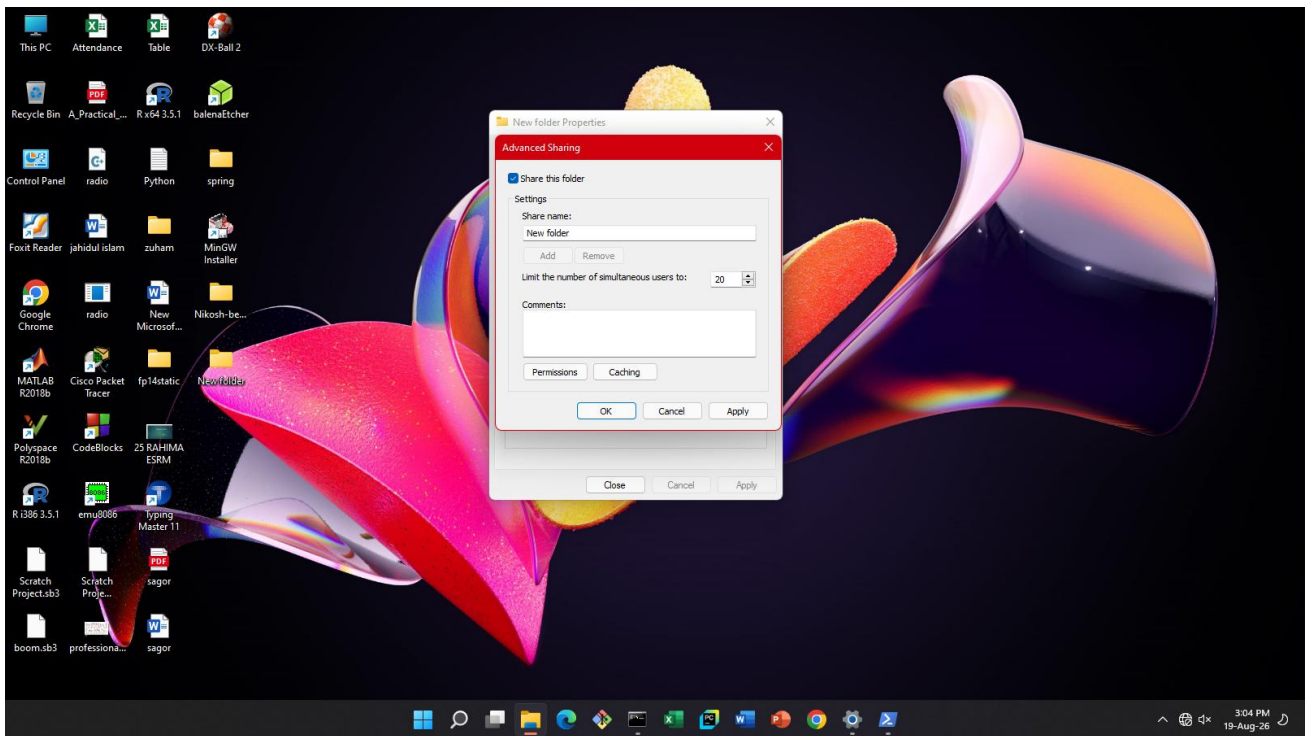
Step 6: Enable File Sharing

On Computer 1:

1. Create or select a folder you want to share
2. Right-click on the folder and select Properties
3. Go to the Sharing tab
4. Click Advanced Sharing
5. Check the box "Share this folder"
6. From "Permissions" control more advanced clearance
7. Click Apply, then OK



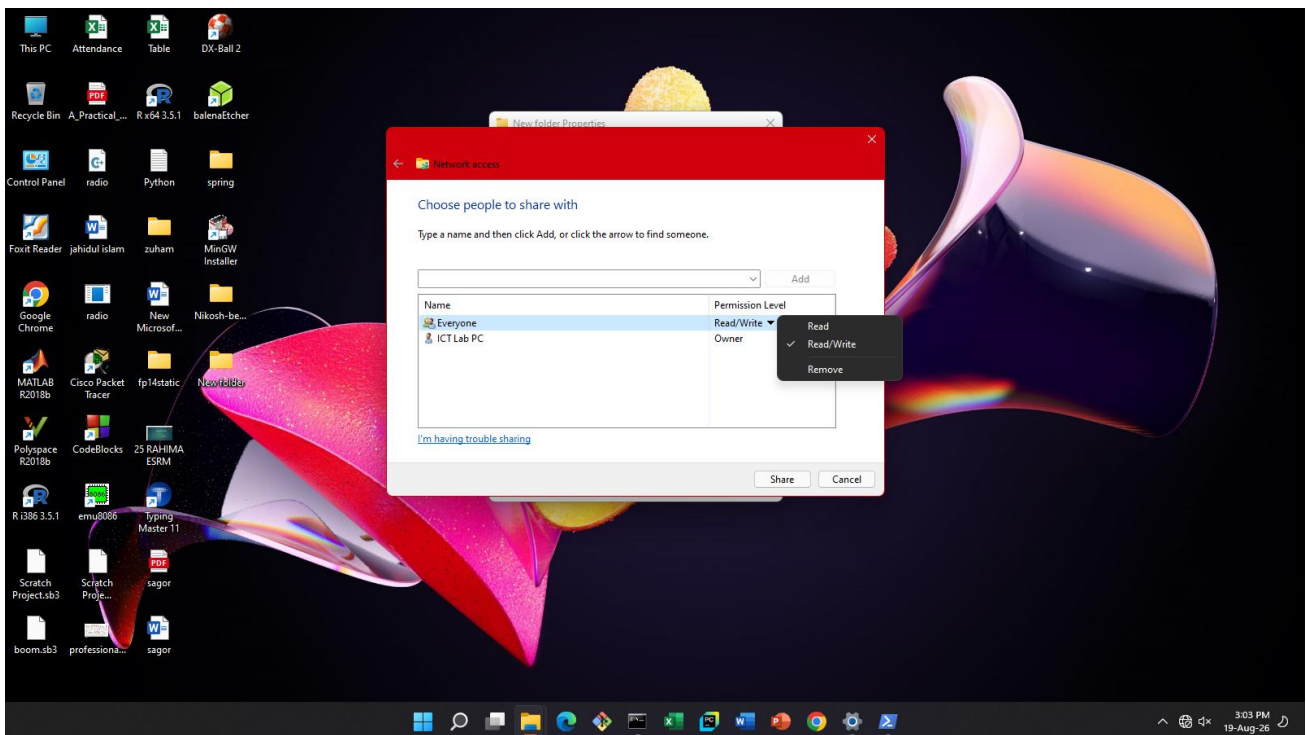




Step 7: Configure Security Access

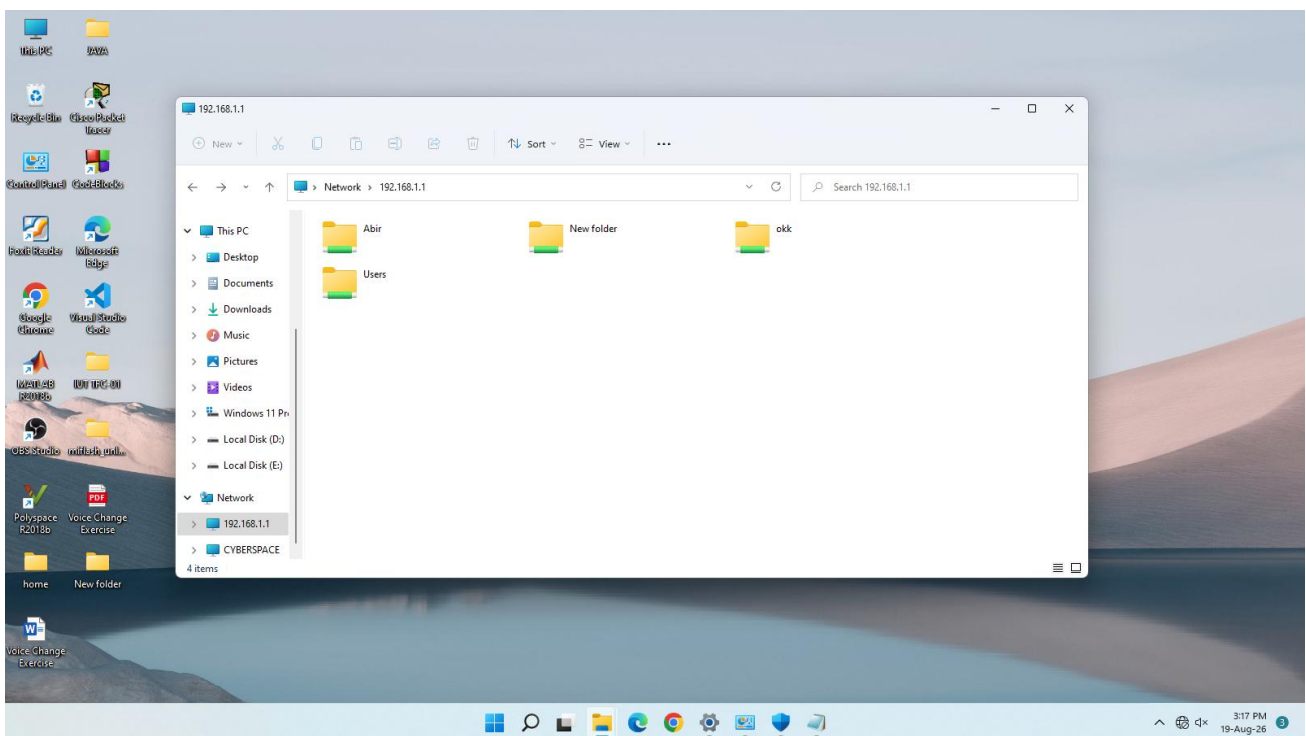
1. In the folder Properties, go to the Security tab
2. Click Edit under "Group or user names"
3. Click Add
4. Type "Everyone" and click Check Names and Click OK
5. Select Everyone and grant full permissions (Read, Write, etc.)

6. Click Apply, then OK



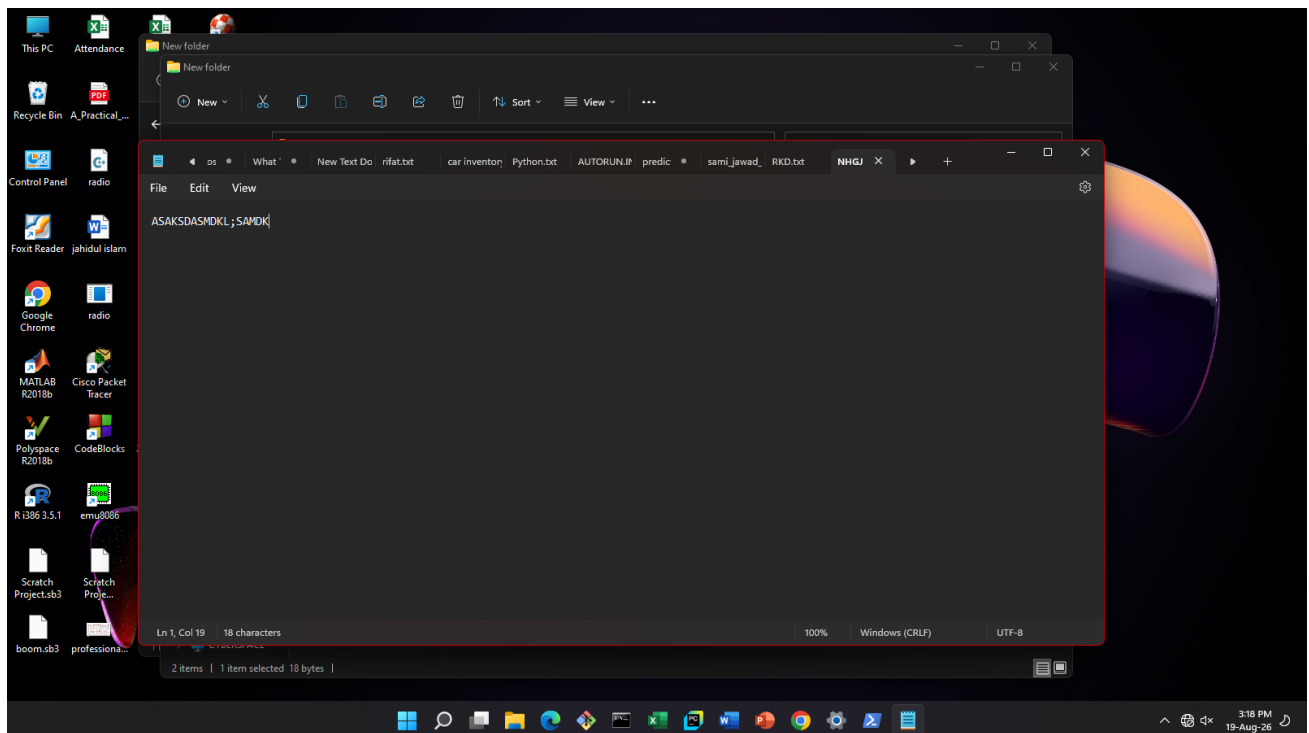
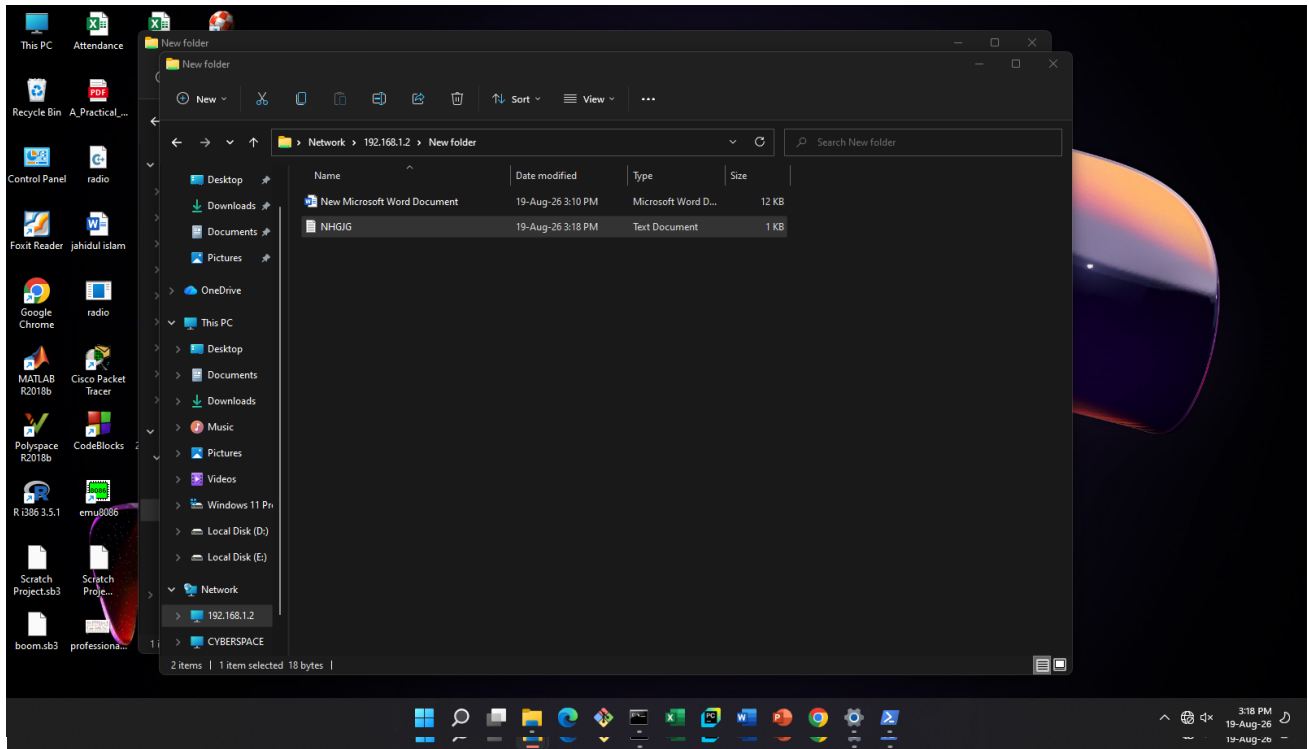
Step 8: Access Shared Folder from Computer 2

1. Open File Explorer on Computer 2
2. In the address bar, type: \\192.168.1.1
3. This will display the shared folder from Computer 1
4. Open the shared folder and verify file transfer capability
5. Create a test file, copy/move files to verify functionality



Step 9: Test File Transfer

1. Copy a file from Computer 1 to the shared folder
2. Access the shared folder from Computer 2 and verify the file is present
3. Create or copy a file from Computer 2 to Computer 1's shared folder
4. Verify bidirectional file transfer is working correctly



Conclusion:

In this experiment, two PCs were successfully connected directly using a Cat6 crossover cable without the need for a switch or hub. Both PCs were manually configured with static IP addresses within the same subnet, and connectivity was verified using the ping command, which confirmed successful data transmission between them. This experiment demonstrated the practical application of a crossover cable in establishing a simple peer-to-peer network, along with the process of manual IP configuration and connectivity testing, which are essential skills in basic computer networking.